

Mohamed Yousry Abdulmajeed

AI Engineer

Qalubia, Egypt

+20 1152052850 yousrymohamed999@gmail.com

linkedin.com/in/mohamed-yousry-0278aa240/ github.com/moYousry14 kaggle.com/m0hamedyousry

Military Status: Exempted

Summary

Artificial Intelligence graduate with hands-on experience building machine learning and deep learning models for real-world challenges, including AI competitions. Proficient in Python, TensorFlow, PyTorch, with a record of developing end-to-end solutions from data cleaning to deployment. Passionate about leveraging intelligent systems to drive automation and solve impactful problems.

AI Competitions & Technical Research

ICMTC 2025 AIC-3 Competition - Team "NeuroScience" 8th Nationwide

June – July 2025

MTC, Cairo, Egypt

[GitHub Link](#)

- **Led team to achieve top rankings:** 7th out of 200+ teams in Phase 1, **8th nationwide** in Phase 2 final round.
- Developed end-to-end pipelines for EEG-based Brain-Computer Interface (BCI) tasks including Motor Imagery (**78% accuracy** with Bidirectional GRU) and SSVEP (**64% accuracy** with Transformer models).
- Enhanced preprocessing with Notch and Bandpass filters, applied real-time fine-tuning using subject-specific adaptation dataset.
- Addressed imbalanced EEG data using advanced techniques: feature engineering, Focal Loss, and temporal modeling.
- Gained hands-on experience with EEG signal acquisition using dry-electrode headsets (Unicorn Hybrid Black).

9th IUGRC 2025 – International Undergraduate Research Conference

July 2025

MTC, Cairo, Egypt

- Participated with research paper: **"Enhancing Patient Care through MediPulse: A Scalable Web and Mobile Health Solution"**
- Integrated AI-powered features including symptom checker with rule-based logic and NLP for initial triage.
- Prompt-tuned **LLaMA-70B** using LangChain for medical question-answering assistant with Groq API deployment.
- Focused on ethical and secure LLM integration within healthcare environments (HIPAA/GDPR compliant).

Zelevstra x AWS ML Ascend Challenge - Team "Imagine Dragons"

June 2025

Solar Panel Efficiency Prediction 84th/1000+ teams

- Developed ML solution for predicting solar panel efficiency using real-time and historical sensor data.
- Engineered domain-specific features and selected Ridge Regression achieving **89.43% custom score** (RMSE 0.1056).
- **Ranked 84th out of 1000+ teams** on the final leaderboard.

Technical Skills

Programming Languages: Python, C++, C#, SQL

Frameworks & Libraries: TensorFlow, scikit-learn, Keras, PyTorch, NumPy, Pandas, Matplotlib

Tools: Streamlit, Gradio, Excel, Git

Concepts: Machine Learning, Deep Learning, NLP, EDA, Model Evaluation, Feature Engineering, BCI

Soft Skills: Problem-solving, Teamwork, Communication, Adaptability

Projects

MediPulse LLM | Healthcare NLP | [GitHub Link](#)

2025

- Deployed a fine-tuned **LLaMA-70B model** for medical question answering using Groq API and LangChain

Emotion Recognition System | LSTM + CNN | [GitHub Link](#)

2025

- Processed audio signals with MFCCs and **hybrid LSTM-CNN architecture** to classify emotions

Fake News Detection | NLP | [GitHub Link](#)

2025

- Trained a **deep learning model** to classify fake vs. real news articles based on text content

Sentiment Analysis | NLP | [GitHub Link](#)

2025

- Built a sentiment classifier using word embeddings and **sequential modeling (LSTM)**

Credit Scoring Model | Supervised ML | [GitHub Link](#)

2025

- Engineered features to train a loan risk classifier; evaluated with **AUC and precision-recall metrics**
- Disease Prediction Models** | *ML/Deep Learning* | [Kaggle Profile](#) **2024-2025**
- Implemented prediction models for **heart disease, Alzheimer’s, diabetes, and breast cancer** using scikit-learn and ANN
- Data Visualization Projects** | *Python & Pandas* | [Kaggle Profile](#) **2024-2025**
- Conducted EDA on **6+ datasets** (Netflix Titles, Egyptian Doctors, Employee Attrition, Used Cars, Café Sales, California Fires)

Internships

- Pathline Academy** **Apr 2025 - May 2025**
Machine Learning Trainee
- Built a **Convolutional Neural Network (CNN)** to classify images of cats and dogs, trained on labeled datasets, and evaluated performance metrics.
 - Trained a **Random Forest model** to analyze feature importance on structured data, identifying key variables influencing predictions; documented approach thoroughly in Jupyter Notebook.
 - Developed an **end-to-end machine learning pipeline** for data preprocessing, model training, hyperparameter tuning, and performance tracking to support scalable deployments.
- NTI & ESE** **Apr 2025 - May 2025**
Artificial Intelligence Trainee
- Completed a **40-hour practical program** covering AI fundamentals, machine learning, deep learning, NLP, data analysis, and ethical considerations.
 - Gained hands-on experience with tools such as **Python, TensorFlow, Power BI, ChatGPT**, and cloud-based AI platforms.
 - Explored applications of AI in **business, healthcare, and government**, including predictive analytics, automation, and data-driven decision making.
- CodeAlpha** **March 2025 - Apr 2025**
Machine Learning Intern
- Built **LSTM-based model** for Speech Emotion Recognition using deep learning on audio signals.
 - Developed a **supervised ML model** for Bank Credit Scoring to predict loan default risk.
- General Authority for Investment & Free Zones (GAFI)** **Aug 2023 – Sep 2023**
Data & IT Intern
- Gained practical experience in **IT infrastructure, network/server operations**, and cybersecurity (Kaspersky).
 - Explored **Power BI** for enterprise-level data analysis and reporting.
 - Studied **DevOps principles**, application lifecycle, and fundamentals of freelancing.

Education

- AkhbarAlyom Academy** **June 2025**
Bachelor of Computer Science – Grade: B+ *6th of October City*

Certifications

- Certificate of participation | MTC **July 2025**
- Image Processing in Python | Datacamp **April 2025**
- AI Diploma | Instant **March 2025**
- AI Training | NTI & ESE **March 2025**
- CSS, JS, Python Complete Course | Udemy **August 2024**
- Summer Training For UNI Students | GAFI **August 2023**

Additional Information

Events & Engagement: Attended TechShift Summit 2025, Egypt Career Summit, and Google Developer Group meetups
Languages: Arabic (Native), English (B2)